Nicholas Williams

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EDUCATION

Ph.D.	2028	Biostatistics, University of California, Berkeley
M.P.H.	2019	Biostatistics, Columbia University
B.A.	2017	Psychology, University of Colorado, Boulder

PROFESSIONAL

Academic

2021-	Columbia University, Mailman School of Public Health Senior Data Analyst III Department of Epidemiology
2019-21	Weill Cornell Medical College Research Biostatistician I Division of Biostatistics, Department of Population Health Sciences
2018	Columbia University, Mailman School of Public Health Research Assistant for Christine Mauro, Ph.D. Department of Biostatistics

Consulting

2021	Bayer AG, Independent Consultant

RESEARCH INTERESTS

Causal inference.

Semiparametric statistics.

Research software engineering.

PUBLICATIONS

Research in statistics

2026	**Williams, N. T.**, Hines, O. J. and Rudolph, K. E. "Riesz representers for the rest of us." *Epidemiology*.

2026 Vo, T.-T., **Williams, N.**, Liu, R., Rudolph, K. E. and Díaz, I. "Recanting twins: Addressing intermediate confounding in mediation analysis." *Statistics in Medicine*.

2024 Gilbert, B., Hoffman, K. L., **Williams, N.**, Rudolph, K. E., Schenck, E. J. and Díaz, I. "Identification and estimation of mediational effects of longitudinal modified treatment policies." *Biostatistics*.

2025 **Williams, N.**, Liu, R., Hoffman, K., Forrest, S., Rudolph, K. and Díaz, I. "Two-stage targeted minimum-loss based estimation for non-negative two-part outcomes." *Statistical Methods in Medical Research*. DOI: 10.1177/09622802251340245

2025 Díaz, I., **Williams, N.**, Hoffman, K. L. and Hejazi, N. S. "Author correction to: 'Causal survival analysis under competing risks using longitudinal modified treatment policies.'" *Lifetime Data Analysis*.

2025 Rudolph, K. E., **Williams, N. T.**, Stuart, E. A. and Díaz, I. "Improving efficiency in transporting average treatment effects." *Biometrika*.

2024 Rudolph, K. E., **Williams, N. T.** and Díaz, I. "Practical causal mediation analysis: Extending nonparametric estimators to accommodate multiple mediators and multiple intermediate confounders." *Biostatistics*.

2024 Rudolph, K. E., **Williams, N.** and Díaz, I. "Using instrumental variables to address unmeasured confounding in causal mediation analysis." *Biometrics*.

2023 Díaz, I., **Williams, N.** and Rudolph, K. E. "Efficient and flexible mediation analysis with time-varying mediators, treatments, and confounders." *Journal of Causal Inference*.

2023 Rudolph, K. E., **Williams, N. T.**, Miles, C. H., Antonelli, J. and Díaz, I. "All models are wrong, but which are useful? Comparing parametric and nonparametric estimation of causal effects in finite samples." *Journal of Causal Inference*.

2023 Rudolph, K. E., **Williams, N.** and Díaz, I. "Efficient and flexible estimation of natural direct and indirect effects under intermediate confounding and monotonicity constraints." *Biometrics*.

2022 **Williams, N.**, Rosenblum, M. and Díaz, I. "Optimizing Precision and Power by Machine Learning in Randomized Trials, with an Application to COVID-19." *Journal of the Royal Statistical Society: Series A*. DOI: 10.1111/rssa.12915

2021 Díaz, I., **Williams, N.**, Hoffman, K. and Schenck, E. "Non-parametric causal effects based on longitudinal modified treatment policies." *Journal of the American Statistical Association - Theory and Methods*. DOI: 10.1080/01621459.2021.1955691

Applied Epidemiology

2026 Cerdá, M., Hung, A.M., Shah, P., Hoffman, K.L., **Williams, N.T.**, Martins, S.S., Doan, L.V., and Rudolph, K.E. "Racial/ethnic differences in prevalence of opioid and non-opioid treatments for pain in Medicaid." *AJE Advances: Research in Epidemiology*.

2026 Rudolph, K.E., Inose, S., **Williams, N.T.**, Díaz, I., Calderon, L., Torres, J.M., and Kioumourtzoglou, M. "Everything all at once: On choosing an estimand for multi-component environmental exposures." *Epidemiology*. DOI: 10.1097/EDE.0000000000002022

2025 Rudolph, K.E., Inose, S., **Williams, N.T.**, Hoffman, K.L., Forrest, S.E., Ross, R.K., Milazzo, F., Diaz, I., Doan, L., Samples, H., Olfson, M., Crystal, S., Cerda, M. and Gao, Y.N. "Mediation of chronic pain and disability on opioid use disorder risk by pain management practices among adult Medicaid patients, 2016-2019." *American Journal of Epidemiology*.

2025 Perry, A., Krawczyk, N., Samples, H., Martins, S.S., Hoffman, K., **Williams, N.T.**, Hung, A., Ross, R., Doan, L., Rudolph, K.E. and Cerda, M. "Opioid Dose, duration, and risk of use disorder in Medicaid patients with musculoskeletal pain." *Pain Medicine*. DOI: 10.1093/pm/pnaf077

2024 Rudolph, K. E., **Williams, N. T.**, Díaz, I., Forrest, S., Hoffman, K. L., Samples, H., Olfson, M., Doan, L., Cerdá, M. and Ross, R. K. "Pain management treatments and opioid use disorder risk in Medicaid patients." *American Journal of Preventive Medicine*.

2024 Hoffman, K. L., Milazzo, F., **Williams, N. T.**, Samples, H., Olfson, M., Díaz, I., Doan, L., Cerdá, M., Crystal, S. and Rudolph, K. E. "Independent and joint contributions of physical disability and chronic pain to incident opioid use disorder and opioid overdose among Medicaid patients." *Psychological Medicine*.

2023 Rudolph, K.E., **Williams, N.**, Díaz, I., Luo, S.X., Rotrosen, J. and Nunes, E.V. "Optimally choosing medication type for patients with opioid use disorder." *American Journal of Epidemiology*. DOI: 10.1093/aje/kwac217

2023 Rudolph, K. E., **Williams, N. T.**, Milazzo, F., Venkataramani, A. and O'Brien, R. "Has the opening of Amazon fulfillment centers affected demand for disability insurance?" *PLOS ONE*.

2022 Rudolph, K. E., **Williams, N. T.**, Goodwin, A. T. S., Shulman, M., Fishman, M., Díaz, I., Luo, S., Rotrosen, J. and Nunes, E. V. "Buprenorphine & methadone dosing strategies to reduce risk of relapse in the treatment of opioid use disorder." *Drug and Alcohol Dependence*.

2022 Hoffman, K.L., Schenck, E.J., Satlin, M.J., Whalen, W., Pan, D., **Williams, N.** and Díaz, I. "Comparison of a target trial emulation framework vs Cox regression to estimate the association of corticosteroids with COVID-19 mortality." *JAMA Network Open*. DOI: 10.1001/jamanetworkopen.2022.34425

2022 Min. J.Y., Simmons, W., Banerjee, S., Wang, F., **Williams, N.**, Zhang, Y., Reese, A.B., Mushlin, A.I. and Flory, J.H. "Association between antidiabetic drug use and the risk of COVID-19 hospitalization in the INSIGHT Clinical Research Network in New York City." *Diabetes, Obesity and Metabolism*. DOI: 10.1111/dom.14704

2022 Min, J.Y., **Williams, N.**, Simmons, W., Banerjee, S., Wang, F., Zhang, Y., Reese, A.B., Mushlin, A.I. and Flory, J.H. "Baselin Haemoglobin A1c and the Risk of COVID-19 Hospitalization Among Patients with Diabetes in the INSIGHT Clinical Research Network." *Diabetic Medicine*. DOI: 10.1111/dme.14815

2022 Hoffman, J.W., Bitar, R.C., **Williams, N.**, Steineman, B.D., Sturnick, D.R., Garrison, G.W., Demetracopoulos, C.A., Drakos, M.C. and O'Malley, M.J. "Orthosis and foot structure affect the fifth metatarsal principal strains during simulated level walking." *The American Journal of Sports Medicine*. DOI: 10.1177/03635465221079652

2022 Punjani, N., Alawamlh, O.H., Kim, S.J., Salter, C.A., Wald, G., Feliciano, M., **Williams, N.**, Dudley, V., and Goldstein, M. "Changes in semen analysis over time: A temporal trend

analysis of 20 years of subfertile non-azoospermic men." *World Journal of Mens Health*. DOI: 10.5534/wjmh.210201

2021 Winston, G., Guadix, S., Lavieri, M., Uribe-Cardenas, R., Kocharian, G., **Williams, N.**, Sholle, E., Grinspan, Z., and Hoffman, C. "Closed-Loop Vagal Nerve Stimulation for Intractable Epilepsy: A Single-Center Experience." *Seizure: European Journal of Epilepsy*. DOI: 10.1016/j.seizure.2021.03.030

2021 Goel, A.R., Bruce, H.A, **Williams, N.** and Alexiades, G. "Long-Term Effects of Hearing Aids on Hearing Ability in Patients with Sensorineural Hearing Loss." *Journal of the American Academy of Audiology*.

2020 Thiesmeyer, J.W., Ullmann, T.M., Greenberg, J., **Williams, N.T.**, Limberg, J., Stefanova, D., Beninato, T., Finnerty, B.M., Vignaud, T., Leclerc, J., Fahey III, T.J., Mirallie, E.M., Brunaud, L. and Zarnegar, R.."Hypertension resolution after adrenalectomy for primary hyperaldosteronism: Which is the best predictive model?" *Surgery*. DOI: 10.1016/j.surg.2020.04.017

2020 Riley, C.A., Zheng, Z., **Williams, N.**, Smith, T.L., Orlandi, R.R. and Tabaee, A. "Concordance of self-reported practice patterns of American Rhinologic Society members with the International Consensus Statement of Allergy and Rhinology: Rhinosinusitis" *International Forum of Allergy and Rhinology*. DOI: 10.1002/alr.22533

2020 Day, J., Conti, M.S., **Williams, N.**, Ellis, S.J., Deland, J.T. and Cody, E.A. "Contribution of First-Tarsometatarsal Joint Fusion to Deformity Correction in the Treatment of Adult-Acquired Flatfoot Deformity." *Foot and Ankle Orthopaedics*. DOI: 10.1177/2473011420927321

2020 Day, J., Kim, J., Conti, M.S., **Williams, N.**, Deland, J.T. and Ellis, S.J. "Outcomes of Idiopathic Flexible Flatfoot Deformity Reconstruction in the Young Patient." *Foot and Ankle Orthopaedics*. DOI: 10.1177/2473011420937985

2020 Cronin, S., Conti, M., **Williams, N.** and Ellis., S.J. "Relationship Between Demographic and Radiographic Characteristics and Second Ray Pathology in Hallux Valgus Patients." *Foot and Ankle Orthopaedics*. DOI: https://doi.org/10.1177/2473011420909088

2019 Hussain, I., Winston, G.M., Goldberg, J., Curri, C., **Williams, N.**, Chazen, J.L., Greenfield, J.P. and Baaj, A.A. "Impact of imaging modality, age, and gender on craniocervical junction angles in adults without structural pathology." *Journal of Craniovertebral Junction and Spine*. PMID: 32089618

Software

2024 **Williams, N.** and Rudolph, K. E. "A drug classification pipeline for Medicaid claims using RxNorm." *arXiv preprint arXiv:2404.01514*.

2023 **Williams, N.** and Díaz, I. "lmtp: An R package for estimating the causal effects of modified treatment policies." *Observational Studies*, 9(2): 103–122.

Commentaries

2025 **Williams, N.**, Hung, A. and Rudolph K. "Re: Don't Let Your Analysis Go to Seed: On the Impact of Random Seed on Machine Learning-based Causal Inference." *Epidemiology*.

DOI: 10.1097/EDE.0000000000001860

2025 **Williams, N.**, Rudolph, K.E. and Díaz, I. "Comment on 'Do we need flexible machine-learning algorithms to assess the effect of long-term exposure to fine particulate matter on mortality?'." *Environmental Epidemiology*. DOI: 10.1097/EE9.0000000000000407

Preprints and manuscripts under review

2026 **Williams, N.**, Schuler, A. "Improving reproducibility by controlling random stability in machine learning based estimation via bagging." *arXiv preprint 2604.17694*. Under review at *Biometrika*.

2026 Díaz, I., **Williams, N.T.**, Morzywolek, P., and Rudolph, K.E. "Modified treatment policies that depend on the natural history of treatment." *arXiv preprint 2605.24167*.

2025 **Williams, N.**, Rudolph, K. and Díaz, I. "Nonparametric estimation of an optimal treatment rule with fused randomized trials and missing effect modifiers." *arXiv preprint 2506.10863*.

2024 Liu, R., **Williams, N. T.**, Rudolph, K. E. and Díaz, I. "General targeted machine learning for modern causal mediation analysis." *arXiv preprint arXiv:2408.14620*.

2024 Susmann, H., **Williams, N. T.**, Rudolph, K. E. and Díaz, I. "Longitudinal generalizations of the average treatment effect on the treated for multi-valued and continuous treatments." *arXiv preprint arXiv:2405.06135*.

SOFTWARE

Maintainer

`lmtp` <R>: Non-parametric causal effects of feasible interventions based on modified treatment policies

`crumble` <R>: Mediation analysis using modified treatment policies, recanting twins, and Riesz representers

`mlr3superlearner` <R>: Super learner fitting and prediction using mlr3

`adjrct` <R>: Doubly-robust and efficient estimators for survival and ordinal outcomes in RCTs without proportional hazards or odds assumptions

Contributer

`broom` <R>: Convert statistical objects into tidy tibbles

INVITED TALKS

2026 Causal Inference Learning Group, Columbia University, New York, NY. "Two attempts at improving finite sample performance of doubly-robust estimators with resampling"

CONFERENCE ACTIVITY

Short courses

"Beyond the ATE." Materials available at beyondtheate.com.

2026 *Society for Epidemiologic Research*, Phoenix, AZ.

2025 *American Causal Inference Conference*, Salt Lake City, UT.

2025 *Society for Epidemiologic Research*, Boston, MA.

2025 *Lifetime Data Science*, Brooklyn, NY.

2025 *American Causal Inference Conference*, Detroit, MI.

2024 *Society for Epidemiologic Research*, Austin, TX.

"Modern Causal Mediation Analysis."

2026 *Society for Epidemiologic Research*, Phoenix, AZ.

2025 *Society for Epidemiologic Research*, Boston, MA.

Conference Presentations

Talks

2026 **Williams, N.T.**, Hung, A., Rudolph, K.E., and Díaz, I. "Modified treatment policies that depend on the natural history of treatment." *European Causal Inference Meeting*, Oxford, United Kingdom.

2026 **Williams, N.T.**, Hoffman, K., Rudolph, K.E., and Díaz, I. "Learning optimal dynamic treatment regimes from longitudinal data." *Society for Epidemiologic Research Midyear Meeting*, Virtual.

2025 **Williams, N.T.**, Rudolph, K.E., and Díaz, I. "Nonparametric estimation of the optimal dynamic treatment rule in fused randomized trials with a missing effect modifier." *American Causal Inference Conference*, Detroit, MI.

Posters

2024 **Williams, N.T.**, Liu, R., Hoffman, K.E., Forrest, S., Rudolph, K.E., and Díaz, I. "Two-step targeted minimum-loss based estimation for non-negative two-part outcomes." *American Causal Inference Conference*, Seattle, WA.

SERVICE

Academic Journal Peer Review

American Journal of Epidemiology

Statistics in Medicine

Journal of Causal Inference

The R Journal

University of California, Berkeley

2026-2027 Co-president, Biostatistics Graduate Student Association

2026-2027 Delegate, Graduate Student Assembly

GRANTS AND AWARDS

Awards and Honors

2017 Graduate with Distinction, University of Colorado at Boulder

Grants and Fellowships

2015 Virginia Summer Undergraduate Research Initiative. Project: "The relationship between cognitively oriented verbs and affect."

TEACHING

University of California, Berkeley

2026 Berkeley Public Health Summer Seminar – Teaching Assistant

Weill Cornell Medicine

Data Science I – Teaching Assistant

Columbia University

Categorical Data Analysis – Lead Teaching Assistant

Research Methods, Quantitative Foundations – Teaching Assistant

PROFESSIONAL AFFILIATIONS

American Statistical Association

Society for Causal Inference

TECHNICAL SKILLS

`R`	expert
`Julia`	intermediate
`SAS`	
`SQL`	
`C++`	novice
`Ruby`	

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